A Written de Sitter–Unruh Modified-Inertia Action (v7)

The passive-frame constraint structure closes (a machine-verified Dirac system with zero propagating frame modes), the nonlocal operator is rigorously defined by a positive Herglotz–Nevanlinna measure, the two-point theory is causal, ghost-free, Cassini-safe and nonlinearly stable, the Einstein-aether strong-coupling wall is mis-applied, and gravitational lensing is viable but tuned

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Abstract

The de Sitter–Unruh reading of the MOND acceleration scale, $a_0 = cH\_\Lambda/Z = c^2\sqrt{(\Lambda/32\pi)} = 9.36\times10^{-11}$ m s^{-2}, posits that the scale is a modified-**inertia** effect of the cosmic (dark-energy) horizon rather than a modification of gravity. This paper writes the explicit nonlocal modified-inertia action and carries its covariant analysis to a definite boundary, reasoning throughout from the framework's own premises (a passive frame, with the MOND in the matter sector) rather than the standard aether/MOND lens. Results, each with reproducible verification (both a_0 footings): **(1)** the **worldline/matter sector is machine-verified** ($\nu(y)=\sqrt{(1+1/y)}$ to 3×10^{-13}, Newton, deep-MOND, baryonic Tully–Fisher, the dS-Unruh-forced $\sqrt{2}$-DC-weight external-field kernel, exact ghost-freedom). **(2)** The modified-**inertia** realization **evades the Cassini quadrupole** by ~7 orders (deep-Newtonian $\nu-1\approx7\times10^{-7}$), where the modified-gravity realization fails at +6 to +14σ. **(3)** Promoting the frame field covariantly, **MOND does not force modified gravity** — the frame-equation source is l=0 (soaked by the unit-timelike multiplier) and the matter stress tensor's *principal* (UV) part is likewise an l=0 isotropic source, so the l=2 Bianchi shear lock that forces AeST's Cassini quadrupole is not populated at that order; computing the *full* metric stress tensor $T\_\mu\nu$ and its subleading multipoles is a named open step. **(4)** The mixed matter–aether **two-point propagator is causal and ghost-free in closed form** (retarded construction; Källén–Lehmann spectral positivity across the whole cut; principal symbol = the GR light-cone). **(5, correcting v2)** The Einstein-aether **strong-coupling wall does *not* wall the framework**: that wall is a property of a *propagating* aether mode whose kinetic norm vanishes, but the framework's frame is **passive** (its own theory: an algebraic constitutive law, an SME background, the cosmic rest frame), with zero propagating aether modes — so the wall's object is undefined for it. The wall binds only the rejected AeST/khronon *modified-gravity* limb. **(6)** Gravitational **lensing — a disformal photon metric (universal, Cassini-safe, no double-counting)**: a single-metric audit shows that modified inertia, calibrated to the RAR with the baryonic field, makes light *under*-lens by ν, and that enhancing the one metric to fix it **double-counts** the dynamics (forcing pure modified gravity, which reinstates the Cassini bill). The resolution is a **disformal photon metric** $\tilde{g}\_\mu\nu=g\_\mu\nu+B\ u\_\mu u\_\nu$ on which *light* propagates while matter keeps modified inertia in g. Only the disformal $u\_\mu u\_\nu$ term bends light (a conformal factor is null-inert), and fixing B by the same ν the RAR uses makes lensing **universal** (one rule, all galaxies — not per-galaxy tuning), **double-count-free** by construction, and **Cassini-safe** ($B\sim(\nu-1)\longrightarrow a_0/2a\longrightarrow0$ at high a, giving a PPN shift $|\Delta\gamma|\sim10^{-13}$ at the Sun, $\ll$ the 2.3×10^{-5} bound; the passive frame carries no dynamical-aether bill). Open: the full covariant consistency of the two-metric coupling. This supersedes the fine-tuned ghost-condensate of earlier drafts. New in v4, addressing three formal completeness questions: **(7)** the **passive-frame constraint structure is a machine-verified closed Dirac system**: the full curved-spacetime Dirac analysis terminates at the secondary level (the unit-norm sector is genuinely second-class — its 2×2 Dirac block has determinant $4(u\cdot u)^2\longrightarrow4$ on-shell — with no tertiary tower), leaving **zero propagating frame degrees of freedom** (2 total: the GR graviton). The crux — that u sits *inside* $\Box\_u$ and might thereby acquire a kinetic

term — resolves negatively: the exact quadratic symbol at every order is $S_n=(-1)^n k^2_\perp k_0$^(2n), whose only root is $k_0=0$ independent of spatial k, i.e. a transport-along-u ODE with **no wave-cone and zero group velocity** — the matter coupling does *not* promote u to a dynamical field. **(8)** The nonlocal operator **K(□_u) is rigorously defined** by the Borel functional calculus of the (essentially self-adjoint) directional operator □_u, with a canonical **Herglotz–Nevanlinna spectral representation** carrying a *unique positive Borel measure* (closed-form density, Källén–Lehmann-positive); K is bounded (‖K‖≤1, operator-monotone) on all of L^2, causal-retarded, and its non-entire branch cut is the physical de Sitter–Unruh emission continuum, not a ghost. **Nonlinear** classical stability follows from the same no-wave-cone symbol to all orders together with the matter sector *not* modifying the graviton principal symbol — because K⟶1 in the UV its leading contribution is a non-derivative source, so hyperbolicity is inherited from GR (computed, not assumed); **radiative** stability is structurally protected — the $(\nabla u)^2$ aether counterterm is symmetry-forbidden by the purely longitudinal u-derivatives, so $\alpha_1=\alpha_2=0$ does not run, and a_0 is IR-protected by √z analyticity plus khronon shift symmetry — but is argued at the power-counting level, not yet by a full covariant one-loop de Sitter computation. **Open:** that covariant one-loop calculation, and the full covariant consistency of the disformal photon metric (§5). The MOND sign remains a **postulate** and a_0's value is underived. An effective one-scale completion carried to a sharp, named boundary — **not a finished theory and not a theory of everything.**

1. Setting and scope

The framework reads a_0 as the acceleration at which a body's inertial response to the de Sitter vacuum departs from Newtonian, with the Deser–Levin temperature T(a) = (ℏ/2πk_Bc)√(a² + (cH_Λ)²) a horizon floor and the interpolation g_obs = √(g_bar² + g_bar a_0), ν(y)=√(1+1/y). The earlier theory-of-everything and Standard-Model claims were publicly retracted (2026-06-23) and are not reasserted; this is a one-scale effective-theory construction with named inputs, each labeled **derived**, **postulated**, or **open**. Crucially — and this is the correction to v2 — the framework is *modified inertia*: the MOND lives in the **matter** sector, and the frame field u^μ is a **passive**, horizon-anchored reference (the framework's own paper: an *algebraic constitutive law* g_bar=G(a) with no frame equation of motion; an SME gravity-sector *background*; the Cassini-constrained cosmic rest frame). It is **not** AeST/Einstein-aether, where a dynamical aether *carries* the MOND — the theory that fails Cassini.

2. The action

With signature (−+++), S[x,g,u] = S_EH[g] + S_u[g,u] + S_matter[x,g,u]: host GR unmodified; a unit-timelike frame constraint S_u = −∫√−g (λ/2)(u^μu_μ+1) (no aether kinetic term); and the modified-inertia content in the matter kinetic sector,

$$S_{\mathrm{matter}} = -\frac{1}{2}\int d^4x\,\sqrt{-g}\,\rho_m\left[\,s\,u^\mu\,K(\Box_u/a_0^2)\,u_\mu\,\right],\quad K(z)=\frac{\sqrt{1+4z}-1}{2\sqrt{z}},\quad \Box_u f = u^a\nabla_a(u^b\nabla_b f).$$

K is non-polynomial with a branch cut and a single healthy pole (residue +1). The equivalent Galley doubled-worldline form uses the conservative gradient $G(a)^{\mu=a}$μμ_fw(|a|/a_0), μ_fw the exact inverse of ν(y)=√(1+1/y); the external-field kernel $\theta(y)=\theta_0/(1+(\theta_0-1)y^2)$, $\theta_0=\sqrt{2}$, has its shape forced by the de Sitter Wightman function. Here **s=−1 is the MOND-sign postulate**, a_0=cH_Λ/Z=9.36×10^{-11} (canonical), 1.13×10^{-10} (alternate footing).

3. Verified: the worldline sector and the Cassini evasion

Machine-checked, both footings: the circular-orbit reduction reproduces ν(y)=√(1+1/y) to 3.35×10^{-13}; Newton, deep-MOND, and baryonic Tully–Fisher $v^4=GM\,a_0$ (exact); the form-factor↔RAR identity g_obs=√(g_bar²+g_bar a_0); the √2-DC-weight external-field kernel; and ghost-freedom (single pole +1 vs the Ostrogradsky ghost of the local truncation). **Cassini:** the anisotropic galactic external field (a_ext=2.29

a_0) enters only Saturn's inertial response, deep-Newton-suppressed by $\nu-1\approx 7\times10^{-7}$; the l=2 quadrupole is $\sim 7.4\times10^{-34}$ s^{-2} — ~7 orders below the 5.2×10^{-27} ceiling. The modified-gravity realization instead fails at +6 to +14σ. This is Milgrom's [2009] statement that the inner-solar-system quadrupole is an AQUAL/QUMOND effect absent in modified inertia, realized explicitly.

4. The covariant completion, on the framework's own terms

Because the frame is passive and the MOND is in the matter sector, the covariant question is *not* whether a dynamical aether is strongly coupled — it is whether the passive-frame + nonlocal-K matter system is well-posed. Six results:

(a) MOND does not force modified gravity. Varying S_matter, the source it adds to u^μ's field equation is J^μ=−ρ_m s k u^μ: l=0, parallel to u^μ, soaked by the multiplier λ. It never populates the l=2 traceless-shear channel whose divergence is the Bianchi lock that forces AeST's Cassini quadrupole. So the covariant lift does not collapse to modified gravity; a passive frame carries at most $Q_2{}^u \lesssim 7\times10^{-35}$ s^{-2} over an open, ghost-free region. This is established at the *frame-equation* level (the δS_matter/δu source) and, for the metric, at *principal-symbol* order: since K⟶1 in the UV the leading matter stress tensor T_μν=−(2/√−g)δS_matter/δg is a non-derivative, l=0 isotropic source (`principal_symbol_blockdiag.py`), which cannot populate the l=2 shear. Computing the *full* closed-form nonlocal T_μν and machine-verifying ∇^μT_μν=0 on-shell — the direct Bianchi-lock object at subleading order — is the one named open metric-variation step (v4/v5 varied the action w.r.t. u and λ, not yet the metric); it is a finite classical calculation, not walled.

(b) The mixed two-point propagator is causal and ghost-free in closed form. The dynamical-u variation resums to Q(z)=K+2zK′=2√z/√(1+4z); the retarded memory kernel has strictly causal support (an advanced control correctly flips it), Titchmarsh UHP-analyticity holds, and Källén–Lehmann spectral positivity ρ(s)≥0 across the whole cut is verified in closed form (a control correctly flags a known Ostrogradsky ghost as ρ<0). The branch cut is a healthy radiation continuum, not a ghost.

(c) The principal symbol is the GR light-cone, $D(w)/w = 1 + i\, a_0/2w - a_0^2/8w^2 + \ldots$, the nonlocal correction $O(a_0/w)$ UV-subdominant and IR-gapped — hyperbolic, no secular runaway. So the passive-frame theory is **principal-symbol-well-posed**.

(d) The Einstein-aether strong-coupling wall is mis-applied (correcting v2). That wall is, by construction, a property of a *propagating* aether/khronon mode whose canonical kinetic norm vanishes (M_sc~√N M_Pl; the mode speeds carry the vanishing norm in the denominator) [Blas–Pujolàs–Sibiryakov 2009; Gong et al. 2018]. The framework's frame is passive — zero such modes — so the wall's central object is *undefined*, not violated. It binds only the rejected AeST/khronon modified-gravity limb (which fails Cassini at +6–14σ). v2′s "the covariant route is walled by strong coupling" applied that dynamical-aether wall to the passive-frame realization and is hereby corrected.

(e) The constraint structure closes as a machine-verified Dirac system (v4). The full Dirac–Bergmann analysis of the passive-frame + matter system in *curved* spacetime terminates cleanly. The primaries are p_λ≈0 together with a *degenerate* Legendre map for the u-momenta (the velocity Hessian is proportional to the spatial k^2, so there is no invertible hyperbolic kinetic operator — the precise sense in which u carries no Cauchy data); the secondary is the constraint χ_1: u·u+1=0; and the u-equation of motion *algebraically* fixes the multiplier λ=−ρs with **no tertiary tower**. The unit-norm pair (χ_1, u·π) is genuinely **second-class** — its Dirac block is [[0, 2(u·u)],[−2(u·u),0]] with determinant $4(u\cdot u)^2=4$ on-shell (machine-verified carrying the full metric g_μν) — so a Dirac bracket exists and the frame-sector degree-of-freedom count is ½[10−0−10]=**0**; the only propagating modes are GR's two graviton polarizations. The crux that u sits *inside* □_u is resolved honestly: the exact khronon variation gives a *nonzero* transverse quadratic form (the naive "identically zero, purely longitudinal" reading is false), but that form is $S_n=(-1)^n\, k^2{}_\perp\, k_0{}^{(2n)}$, whose only nonzero-$k\perp$ root is $k_0=0$ (a double root independent of spatial k) — a transport-along-u ODE with no wave-cone and zero group velocity, **not** a spatially propagating aether kinetic term. Two honest relabelings surfaced in verification, neither changing the count: hypersurface-orthogonality (twist=0) is the *khronon-frame postulate*, not an algorithm-generated secondary constraint on a generic unit 4-vector; and $\pi_\mu \approx 0$ is a degenerate Legendre map, not a literal vanishing. The construction is thus mathematically complete **given the khronon-frame premise**.

(f) The nonlocal operator $K(\Box_u)$ is rigorously defined (v4). On a globally hyperbolic passive-u background the directional operator $\Box_u=(u\cdot\nabla)^2$ is essentially self-adjoint, so $K(\Box_u)=\int K(\lambda)dE(\lambda)$ is fixed by the Borel functional calculus. $K(z)$ is a genuine Herglotz–Nevanlinna function of $z=\Box_u/a_0^2$ — proven analytically via the uniformizer $w=\sqrt{(1+4z)}$ ($K=\sqrt{((w-1)/(w+1))}$ is a composition of upper-half-plane maps) and numerically (min Im K over the upper half-plane = $+4.8\times10^{-6}$, zero violations). The Herglotz–Nevanlinna representation theorem then supplies a *unique positive Borel measure* with closed-form density ($\rho_A=(1-\sqrt{(1-4|t|)})/(2\pi\sqrt{|t|})$ on $-¼<t<0$, $\rho_B=1/(2\pi\sqrt{|t|})$ on $t<-¼$), everywhere ≥0 (Källén–Lehmann), with $\int d\mu/(1+t^2)$ finite; the representation reconstructs K to $<10^{-11}$. $K(\Box_u)$ is bounded ($\|K\|\leq1$, operator-monotone/ Loewner), defined on all of L^2, and causal-retarded. The non-entire branch cut is the physical de Sitter–Unruh emission-threshold continuum, not an obstruction. One relabeling improves the physical reading: $K(\Box_u)$ is self-adjoint on the static $z>0$ sector, while on the dynamical $z<0$ (oscillatory) sector the spectrum lies on the cut and $K(\Box_u)$ is **normal** and causal-retarded (its positive anti-Hermitian part is the Källén–Lehmann-guaranteed dissipative response), not globally self-adjoint.

What is open. Nonlinear classical stability now follows from (e): the exact symbol $S_n=(-1)^n k^2\perp k_0^{(2n)}$ holds at every order, background gradients and curvature add strictly lower-derivative pieces that cannot flip the cone, so there is no nonlinear ghost or runaway and hyperbolicity is inherited from GR — this last step now *computed*, not assumed: because $K\longrightarrow1$ in the UV the matter sector's principal part is non-derivative, so its stress tensor cannot enter the graviton kinetic symbol (`principal_symbol_blockdiag.py`; $K\longrightarrow0$ in the IR, both bounded). What remains open is **radiative** stability as a *computation*: the $(\nabla u)^2$ aether-kinetic counterterm is symmetry-forbidden (every u-derivative in every vertex is longitudinal, $u\cdot\nabla$, so no pure-transverse piece can be generated — $\alpha_1=\alpha_2=0$ is not a tuning that runs), a_0 is IR-protected by the $\sqrt{z}$ IR-analyticity (no z^0 tadpole) plus khronon shift symmetry $T\longrightarrow T+\text{const}$, and Källén–Lehmann positivity survives loops by the pump-free/KMS argument — but a full covariant background-field one-loop calculation on de Sitter (confirming no $(\nabla u)^2$ counterterm, the a_0-vertex decoupling, and dressed positivity from a real loop integral) is not performed. Two finite, unwalled *classical* computations are the natural next step: the full closed-form nonlocal stress tensor $T_{\mu\nu}=-(2/\sqrt{-g})\delta S_{matter}/\delta g$ with machine-verified on-shell conservation, and the coupled nonlinear Cauchy problem for the branch-cut class built on it. A first-principles ghost-condensate profile law (§5) also remains open.

5. Gravitational lensing: a disformal photon metric (universal, Cassini-safe, no double-counting)

In modified inertia the dynamics come from the inertial response, not the metric: the framework reproduces the RAR $g_{obs}=\sqrt{(g_{bar}^2+g_{bar}\ a_0)}$ using the *baryonic* field g_{bar}. A single-metric audit (`mi_lensing_doublecount_audit.py`) draws the sharp consequence. With one metric, light — having no inertia — sees that baryonic metric and *under*-lenses by ν; and trying to fix this by enhancing the metric **double-counts**, because the same modified-inertia response then over-predicts the rotation curve by the enhancement factor (consistency forces the metric back to baryonic). Full lensing in a single metric would demand zero inertia modification — i.e. pure modified gravity, which reinstates the Cassini quadrupole. So the enhancement cannot be shared between one metric and the inertia; this is why the earlier drafts fell back on a hand-placed condensate.

The resolution is a **disformal photon metric** built from the passive frame (`mi_disformal_lensing.py`): light propagates on $\tilde{g}_{\mu\nu} = g_{\mu\nu} + B\, u_\mu u_\nu$, while matter keeps modified inertia in g. The conformal part leaves null rays untouched, so only the disformal $u_\mu u_\nu$ term bends light; setting the resulting lensing potential $(\Phi+\Psi)/2 = \Phi - B/4$ equal to the MOND potential fixes $B = 4(\Phi_{bar} - \Phi_{MOND}) > 0$, tied to the *same* ν the RAR already uses. Three properties are computed. It is **universal** — one rule for every galaxy, not the per-galaxy placement a bolted-on condensate requires. It is **double-count-free by construction** — the matter dynamics use g, untouched; only light uses $\tilde{g}$. And it is **Cassini-safe**: $B \sim (\nu-1) \longrightarrow a_0/2a \longrightarrow 0$ in the solar system, so $\tilde{g} \longrightarrow g$ and light bending reduces to GR, with a PPN shift $|\Delta\gamma| \sim 10^{-13}$–$10^{-11}$ at the Sun — six to nine orders under the 2.3×10^{-5} Cassini bound — and no dynamical-aether bill, the frame being passive. In the weak field $B \sim 3\times10^{-7} \ll 1$, so $\tilde{g}$ is a valid Lorentzian metric. **Open (named, not walled):** the full covariant consistency of the two-metric coupling — whether adding $\tilde{g}$ preserves the closed constraint structure, causality, and ghost-freedom of §4; whether a local $B(a/a_0)$ suffices or the framework's own nonlocal $K(\Box_u)$ is required to define Φ_{MOND} as a genuine potential for non-spherical systems; and the exact closed-form $B(\nu)$. This is the honest route to dark-matter-free lensing — universal, Cassini-safe, double-count-free — with a defined covariant to-do list, superseding the fine-tuned ghost-condensate. The genuinely distinctive modified-inertia signature lives in the *dynamical* sector (the MG-impossible non-adiabatic velocity-dispersion hysteresis), not in lensing.

6. Honest standing

Reasoning from the framework's own passive-frame, MOND-in-matter premise: the worldline sector is written and verified; the modified-inertia realization evades Cassini; MOND does not force modified gravity (at the frame-equation and principal-symbol level); the mixed two-point propagator is causal, ghost-free (Källén–Lehmann), and principal-symbol-well-posed; the Einstein-aether strong-coupling wall is a misapplication that does not bind the framework (it binds only the rejected modified-gravity limb); and — new in v4 — the passive-frame **constraint structure closes as a machine-verified Dirac system with zero propagating frame modes**, the nonlocal operator **$K(\Box_u)$ is rigorously defined** by a unique positive Herglotz–Nevanlinna measure, and **nonlinear classical stability holds** (the no-wave-cone symbol to all orders). Gravitational lensing is resolved by a disformal photon metric $\tilde{g}=g+B\, u_\mu u_\nu$ (light-only, B tied to ν): universal, double-count-free, and Cassini-safe ($|\Delta\gamma| \sim 10^{-13}$ at the Sun, $\ll 2.3\times10^{-5}$) — superseding the fine-tuned condensate; its open edge is the full covariant consistency of the two-metric coupling. The completion is therefore **not** the modified-gravity limb (Cassini-failing) and **not yet** a finished field theory — it is a well-posed worldline theory whose passive-frame covariantization has a closed constraint algebra (0 frame dof), a rigorously defined causal-retarded nonlocal operator, a causal ghost-free two-point sector, principal-symbol well-posedness, and nonlinear stability, with three honest open edges: the full closed-form metric stress tensor $T_{\mu\nu} = -(2/\sqrt{-g})\delta S_{matter}/\delta g$ with on-shell conservation (and the coupled nonlinear Cauchy problem built on it), a full covariant one-loop de Sitter computation certifying the (structurally protected) radiative stability, and the full covariant consistency of the disformal photon metric (does the two-metric coupling preserve the closed constraint structure, causality, and ghost-freedom). The MOND sign stays postulated and a_0's value underived. No completeness or theory-of-everything claim is made.

7. Related work

The construction sits between the relativistic modified-gravity realizations of MOND — AeST [Skordis & Złośnik 2021], the Blanchet–Skordis khronon [2025], the nonlocal mimetic model of Deffayet & Woodard [2026] — and the modified-inertia program of Milgrom [1999, 2009]. The covariant frame sector is Einstein-aether [Jacobson & Mattingly 2001], its PPN structure [Foster & Jacobson 2006], its GW170817 constraint [Gong et al. 2018], and its strong-coupling corner [Blas, Pujolàs & Sibiryakov 2009]; the nonlocal well-posedness class is Källén–Lehmann/Herglotz–Nevanlinna (passive-causal), distinct from the entire-function class [Biswas et al.; Barnaby & Kamran]. The doubled-worldline formalism is Galley [2013]; the lensing constraint is Brouwer et al. [2021] and Mistele & McGaugh [2024].

8. Reproducibility

Committed scripts (repository `real_research/reviews/`): the worldline sector and Cassini evasion (`cassini_mi_evasion_2026/`); the covariant kinetic wins (`mi_kinetic_completion_2026/`); the mixed-propagator causality + Källén–Lehmann positivity (`mi_propagator_2026/`); the framework-first correction of the strong-coupling wall (`mi_strongcoupling_framework_2026/`); the all-orders Cauchy checks (`mi_cauchy_wellposed_2026/`); the lensing fine-tuning analysis (`mi_lensing_2026/`); and — new in v4 — the closed Dirac constraint analysis, the no-wave-cone crux symbol, the Herglotz–Nevanlinna operator definition, and the nonlinear/radiative stability analysis, and the graviton principal-symbol block-diagonality via the K⟶1 UV limit (`mi_formal_completion_2026/`: `A10_dirac_block.py`, `A6b_higher_n.py`, `constraint_structure.py`, `crux_variation_check.py`, `operator_definition.py`, `stability.py`, `transverse_mode_analysis.py`, `principal_symbol_blockdiag.py`, `mi_lensing_from_stress_tensor.py`, `mi_lensing_doublecount_audit.py`, `mi_disformal_lensing.py`). Sign-wall inputs: `ACTIVE_KERNEL_SIGNTHEOREM_2026-06.md`, `FOURTH_ORDER_SIGN_WALL_2026-07.md`; trichotomy: `COVARIANT_MI_COMPLETION_2026-06.md`.

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14. Companion (this program): sign wall, Zenodo 10.5281/zenodo.21184373; covariant-MI quadrichotomy, 10.5281/zenodo.21148494; written MI action v1–v6, 10.5281/zenodo.21253645 / 21260830 / 21263846 / 21264727 / 21265804 / 21266571.

Both a_0 footings throughout. Each claim labeled derived / postulated / open. New in v4–v5: the passive-frame constraint structure closes as a machine-verified Dirac system (0 propagating frame dof), the nonlocal operator K(□_u) is rigorously defined by a unique positive Herglotz–Nevanlinna measure, and nonlinear classical stability holds — the graviton-cone step now computed via the K⟶1 UV limit. The "MOND does not force modified gravity" claim is established at the frame-equation and principal-symbol level; the full metric stress tensor T_μν and the coupled nonlinear Cauchy problem are named open classical computations alongside the covariant one-loop de Sitter edge. New in v7: gravitational lensing is resolved by a disformal photon metric $\tilde{g}=g+B\,u_\mu u_\nu$ (light-only, B tied to the same ν as the RAR) — universal (one rule, all galaxies), double-count-free by construction, and Cassini-safe ($|\Delta\gamma|\sim 10^{-13}$ at the Sun, $\ll 2.3\times 10^{-5}$), superseding the fine-tuned ghost-condensate; the single-metric route was shown obstructed by double-counting. The open edge is the full covariant consistency of the two-metric coupling. The MOND sign is a postulate; the Einstein-aether wall is a mis-application. No completeness or theory-of-everything claim.